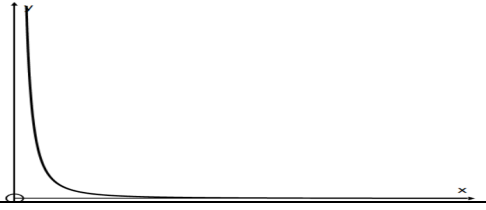
Question			Answer	Marks	AOs		Guidance
11	i		$y = kx^n$	**B1** **[1]**	**3.3**	Allow any letters used for constant of proportionality and power.	
	ii		$\ln y = \ln\left(kx^n\right)$ $\ln y = \ln k + \ln\left(x^n\right) = \ln k + n\ln x$	**M1** **E1** **[2]**	**2.1** **2.1**	Taking natural logs of both sides and one correct use of laws of logs used. Convincing argument. **AG**	
	iii		<table><tr><td></td><td>ln$x$</td><td>ln$y$</td></tr><tr><td>Mercury</td><td>-1.179</td><td>9.575</td></tr><tr><td>Jupiter</td><td>1.599</td><td>4.022</td></tr></table>	**B1** **B1** **[2]**	**1.1b** **1.1b**	At least 2 correct values All correct and to 4sf	
	iv		EITHER $b = \dfrac{9.575 - 4.022}{-1.179 - 1.599} = -1.999$ (-2.00 to 3sf) $a = 7.218$	**M1** **A1** **A1** **[3]**	**1.1a** **3.1a** **1.1b**	using gradient formula Allow -2 a correct to at least 2 sf	These values could be found using the calculator STATS mode, so allow without working
			OR $9.575 = a - 1.179b$ $4.022 = a + 1.599b$ Giving $a = 7.218$ $b = -1.999$ (-2.00 to 3sf)	**M1** **A1** **A1** **[3]**		Setting up pair of equations by substitution of their values. Allow one slip. a correct to at least 2 sf b correct to at least 2 sf	Simultaneous equations can be solved using calcuator
	v		$y = 1363x^{-2.00}$	**B1** **B1** **[2]**	**2.2a** **3.3**	FT their equation in (i) awrt 1300 or 1400, or $e^{7.2}$ or better. FT their a Allow for x^{-2} or better. FT their b	
	vi			**B1** **B1** **[2]**	**1.2** **1.1b**	Appropriate curve with at least one horizontal asymptote or vertical asymptote shown Both asymptotes correct Ignore $x < 0$ if shown	FT their equation in (v) provided their function is a decreasing function
	vii		Earth $x = 1$, $y = 1363 \times 1^{-2} = 1360$ W m^{-2} (3sf)	**B1** **[1]**	**3.4**	FT their (v)	

Question	Answer	Marks	AOs		Guidance
	SCHEME FOR CANDIDATES USING LOG BASE 10				
ii		**SC1**		Correct use of log instead of ln and no other error	
iii	<table><tr><td></td><td>ln$x$</td><td>ln$y$</td></tr><tr><td>Mercury</td><td>-0.5122</td><td>4.158</td></tr><tr><td>Jupiter</td><td>0.6946</td><td>1.747</td></tr></table>	**B0** **B1**		All correct. Must be 4 sf	
iv	As in main scheme a = 3.1336 b = -1.999	**M1** **A1** **A1**			
v	$y = 1360x^{-2}$	**B1** **B1**		FT their equation in (i) awrt 1300 or 1400, or $e^{3.1}$ (= 22.2), $10^{3.1}$ or better. FT their a allow for x^{-2} or better. FT their b	
vii	Earth $x = 1$, $y = 1363 \times 1^{-2} = 1360$ W m^{-2} (3sf)	**B1**		FT their (v)	